
MA 601: Real and Functional Analysis

Course Code	MA 601
Course Name	Real and Functional Analysis
L-T-P-C	2-1-0-3
Prerequisites	Basic Analysis
Intended for	Masters/Pre. Ph.D
Distribution	Elective
Approval	1st Adhoc Senate (Apr 2011)

Course Contents

- **Metric spaces** [Lectures]

Open sets, Closed sets, Continuous functions, Completeness, Cantor intersection theorem, Baire category theorem, Compactness, Totally boundedness, finite intersection property. Definition and existence of Riemann-Stieltjes integral, Properties of the integral, Differentiation and integration. Uniform convergence, Uniform convergence and continuity, Uniform convergence and integration, Uniform convergence and differentiation.

- **Normed linear spaces** [Lectures]

Normed linear spaces, Riesz lemma, characterization of finite dimensional spaces, Banach spaces. Bounded linear maps on normed linear spaces: Examples, linear map on finite dimensional spaces, finite dimensional spaces are isomorphic, operator norm. Hahn-Banach theorems. Uniform boundedness principle, closed graph theorem, open mapping theorem, inner product spaces, orthonormal set, Gram-Schmidt orthonormalization orthonormal basis, orthonormal complements.

Text Books

1. J. Conway, **A Course in Functional Analysis**, 2nd Edition, Springer.
2. W. Rudin, **Principles of Mathematical Analysis**, McGraw-Hill, 1986.
3. N. L. Carother, **Real Analysis**, Cambridge University Press, 2000.

References

1. E.T. Copson, **Metric Spaces**, Cambridge University Press, 1968.
2. Claude W. Burtil, John R. Knudsen, **Real Variables**, Holt, 1969. t, Reinhart and Winston
3. Tom M. Apostol, **Mathematical Analysis**, Addison Wesley, 1974.
4. BLiamlmitoehda, (n1 V9.9 L6i)m aye, **Functional Analysis**, 1st Edition, New Age International